

Coverage: Porsche AG (ETR: P911) vs. Ferrari N.V. (BIT/NYSE: RACE) | **Classification:** Automotive & Luxury Goods — Relative Value | **Sample:** 1,008 trading days, Sept 2022 – Sept 2026

Full Quantitative Model & Python Source Code: [\[GITHUB REPOSITORY LINK — e.g. github.com/gayathritvinod/porsche-ferrari-quant-framework\]](#)

Rejecting a Naive Mean-Reversion Trade

A four-part econometric and fundamental test battery on the P911/RACE price ratio — and why the “obvious” reversion trade would have failed.

Verdict: Structural Value Trap, Not Statistical Arbitrage. The P911/RACE price ratio spent two years (2022–2024) oscillating around $\sim 0.25\times$ — tempting a straightforward pairs trade: buy Porsche, sell Ferrari, wait for reversion. Before recommending that trade, I stress-tested it with an institutional-grade battery: stationarity, structural break, reversion speed, and a historical backtest. **Every independent test rejects it.** Recommendation: **NO ACTION** on a reversion trade, with explicit re-test triggers set below.

The Four-Test Verdict

Test	What It Asks	Result	Verdict
Stationarity(Augmented Dickey-Fuller)	Does the ratio actually pull back to a mean, or just wander?	$p = 0.6717$	Fails to reject a random walk — no statistical pull-back force
Structural Break(Chow Test)	Did the regime the “0.25x” anchor came from actually end?	$F = 1,081.01$, $p < 0.0001$ (break: June 2024)	Break confirmed — the historical anchor is statistically invalid
Reversion Speed(Ornstein-Uhlenbeck)	If it did revert, how fast — and how confident are we?	$\kappa = 0.40/\text{yr}$; half-life ≈ 435 days; 95% CI spans zero	Point estimate plausible; not statistically resolved
Historical Backtest(29 episodes)	Has this exact setup actually paid off before?	13.8% 12-month hit rate; avg. return $+1.6\%$	Worse than a coin flip — the pattern hasn't recurred

Independent Fundamental Cross-Check

A bottom-up EV/EBITDA scenario engine (bear/base/bull, probability-weighted) corroborates the statistical read using none of the same data: model-implied fair ratio of **0.076**, against a current market ratio of ≈ 0.125 — both sit well below the original 0.25 target, not above it. The business explains why: Porsche is mid-restructuring (*EV margin dilution, $\sim 9,000$ disclosed job cuts through 2035, China demand deterioration*), while Ferrari's order book *covers production into 2027 with EBITDA margin above 39%*. This is a business-model divergence, not a temporary mispricing.

Current Regime Z-Score	Annualized Tracking Error	Information Ratio
-0.38 (no actionable stretch today)	33.2%	1.16 (caveated — see full report)

Recommendation & Monitoring Triggers

NO ACTION on a reversion trade. Re-test triggers: quarterly κ re-estimation; Porsche Capital Markets Day (Oct 7, 2026) and Q3 statement (Oct 27, 2026); Ferrari Q3 results (Nov 3, 2026). Full monitoring framework and half-life breach rule in the linked report.

Methodology: Python · statsmodels (ADF) · hand-built Chow F-test · Ornstein-Uhlenbeck fit via AR(1) OLS with a 1,000-draw bootstrap · EV/EBITDA scenario engine · historical signal backtesting. Fully reproducible via a frozen, version-controlled data snapshot (see repository).